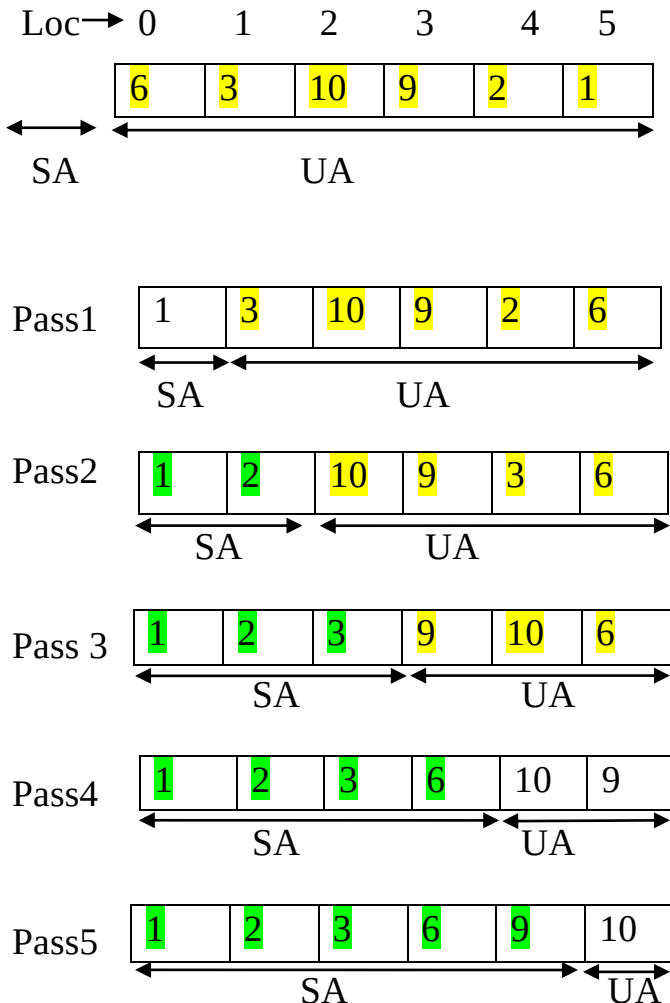


SELECTION SORT ALGORITHM:-

Selection sort algorithm array is divided in to two sub arrays or lists one is sorted sub array or list and other is unsorted sub array or list, initially we assume the sorted sub array is empty so by taking the smallest element from an unsorted array and bringing it to the **beginning of sorted array** and keep on doing this until the unsorted array is shifted to sorted array. Time complexity of algorithm is $O(n^2)$.

Example :- (For N=6)



```
int Selection sort (int arr [])  
for (i=0; i < n-1; i++)
```

```
{  
  int min= i; int temp;  
  for( j = i+1; j < n ; j++)  
  {  
    if ( a [ j ] < a [ min ] )  
    {  
      min = j;  
    }  
  }  
  if (min != i)  
  {  
    temp = arr[ i ]  
    arr[ i ] = arr[min]  
    arr[min] = temp;  
  }  
}
```

Explanation

min = 5

j = 5

if (a[j] < a[min])

if(a[5] < a[4])

Selection sort Algorithm	Time Complexity
{	
int minindex;	1
int temp;	1
for (int i=0; i < n-1;i++)	$1+N-1+1+N-1 = 2N$
minindex= i;	N-1
for(int j = i+1; j < n ; j++)	$N-1+x+1+x$ (assume $N-1=x$)
{	
if (a [j] < a [minindex])	3x
{	
minindex = j;	x
}	
}	
{	
temp = arr[i]	$2(N-1)$
arr[i] = arr[minindex]	$2(N-1)$
arr[minindex] = temp;	$2(N-1)$
}	
return arr;	1
}	
	$T(N) = 1+1+2N+N-1+N-1+x+1+x+3x+x+2(N-1)+ 2(N-1)+ 2(N-1)+1$ $= 6x+10N-4$
	$\therefore X= 1 + 2+3 +\dots\dots(N-1)$ in Arithmetic progression
	Considering AP $1+2+3+\dots\dots k = K(K+1)/2$
	$X= (N-1) (N-1+1)/2$
	$6N^2/2 - 6N/2 +10N-4$
	$T(N) = 3N^2 - 3N + 10N - 4$
	$T(N) = 3N^2 + 7N -4$
	Considering big O notation and ignoring constant we consider the term of higher growth.
	$T(N) = O(N^2)$.